Predictive Approaches to Treatment Effect Heterogeneity

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Background

• Relying on average trial effects ignores that patients differ in demographics, disease severity, and behaviors - termed patient Heterogeneity (HTE) [1], crucially, these factors may modify the effectiveness of treatment.

• Traditional subgroup analyses (one variable at a time) have low power, high false positives, and aren't truly patient-centred.

• Predictive Approaches to Treatment effect Heterogeneity (PATH) model risk and treatment effects to estimate individual patient outcomes across treatments [2].

Objectives

1. Implement and evaluate predictive HTE models across multiple clinical datasets
2. Compare risk-based and effects-based approaches in practice
3. Assess the clinical relevance and reporting implications of PATH analyses

Data Sources

Analyses are based on published clinical trial data from the GUSTO-I trial [3] and the Digitalis Investigation Group trial [4].

• The GUSTO trial was an international, randomized clinical trial comparing four thrombolytic strategies in 41,021 patients with acute myocardial infarction, conducted from 1989–1993, showing that accelerated t-PA with intravenous heparin reduced 30-day mortality compared with streptokinase-based regimens.

• The DIG trial was a randomized, double-blind, multicentre clinical trial of digoxin vs placebo in 6,800 heart-failure patients conducted from 1991–1995 with a mean 37-month follow-up, showing no reduction in overall mortality but fewer hospitalizations for worsening heart failure with digoxin.

Early Results - Gusto

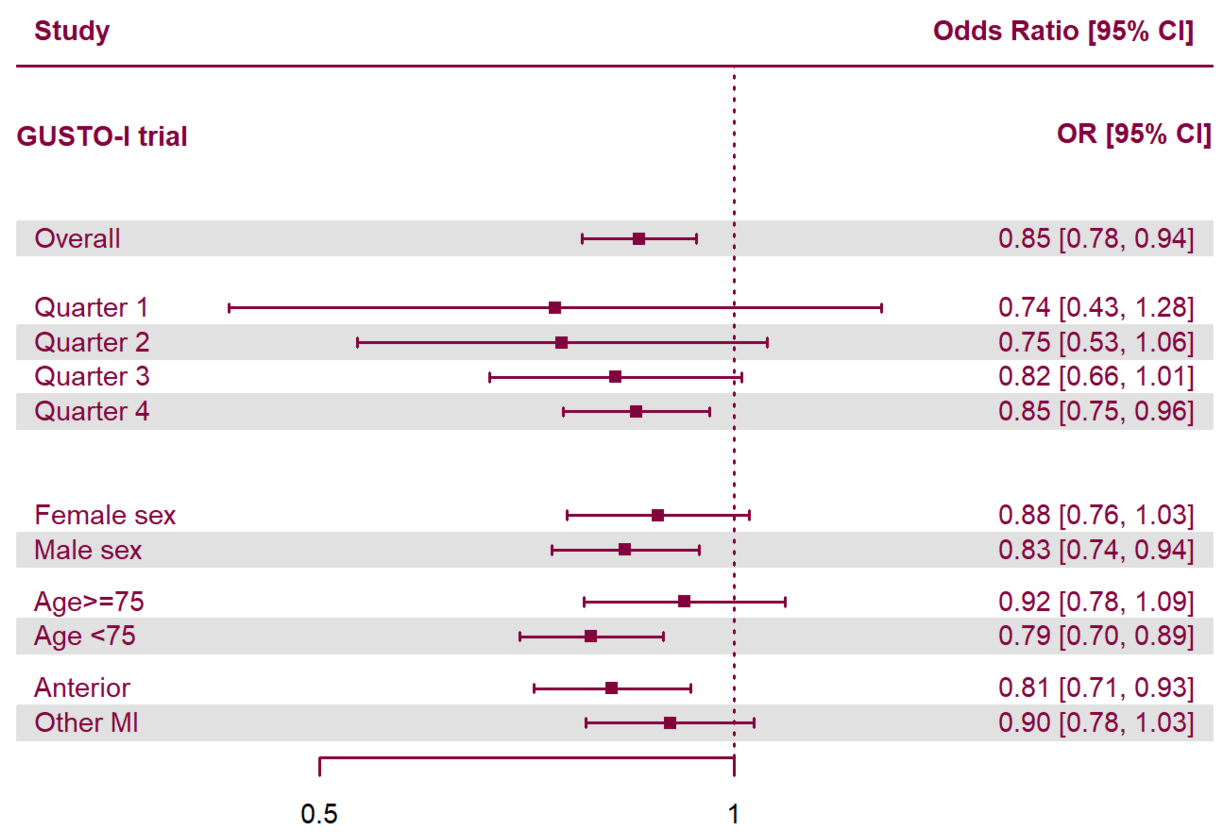

Figure 1: Forest plot of treatment heterogeneity

Above is a Forest Plot which display the PATH (risk-based) subgroups alongside traditional subgroups (Age, Sex, Location of MI). Low risk according to the linear predictor (lp<-2.36) implies low benefit (<1%); higher risk, higher benefit (>2%).

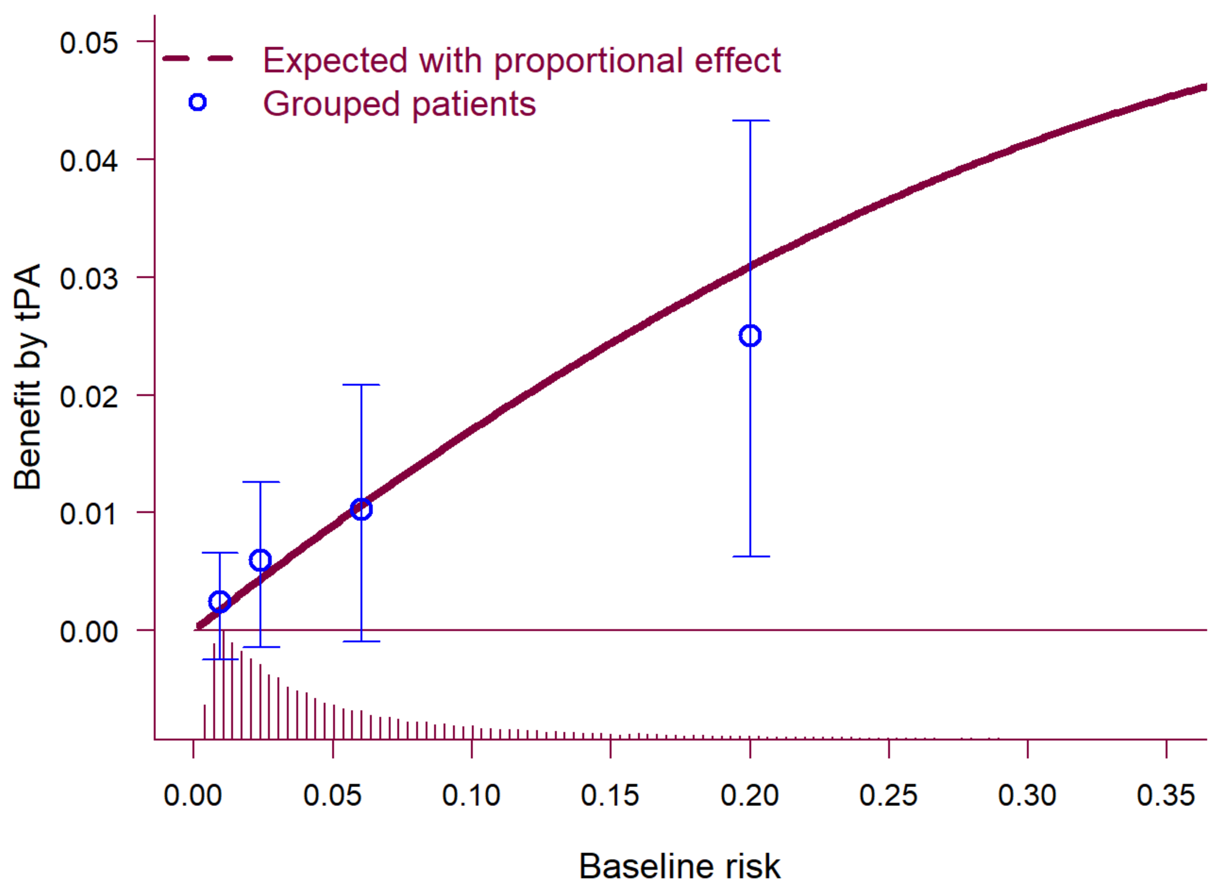

Figure 2: Absolute treatment benefit increases with baseline risk

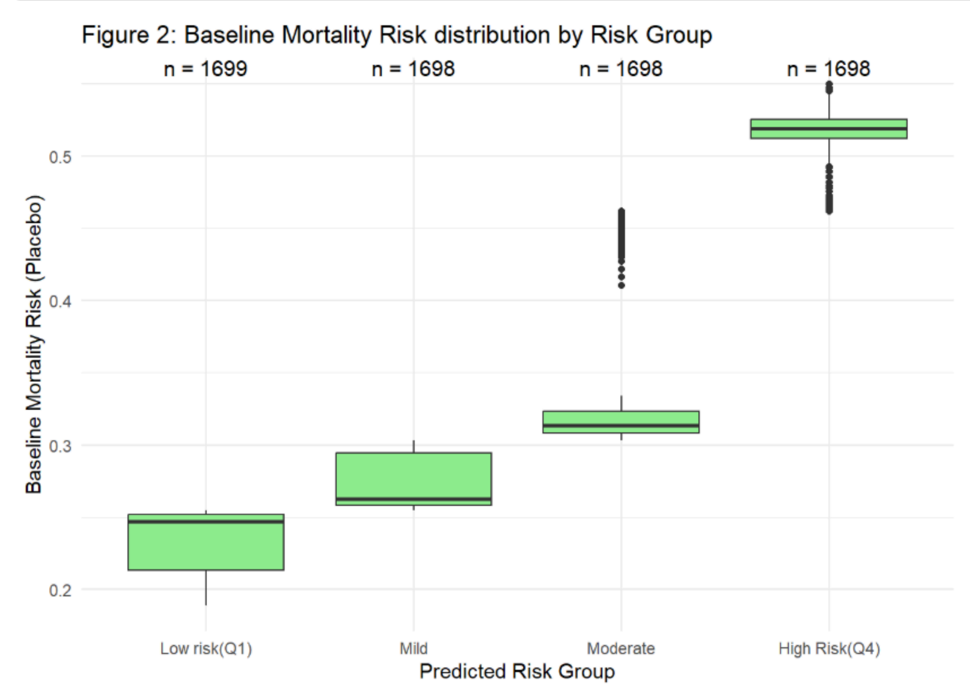

Figure 3: Baseline Risk

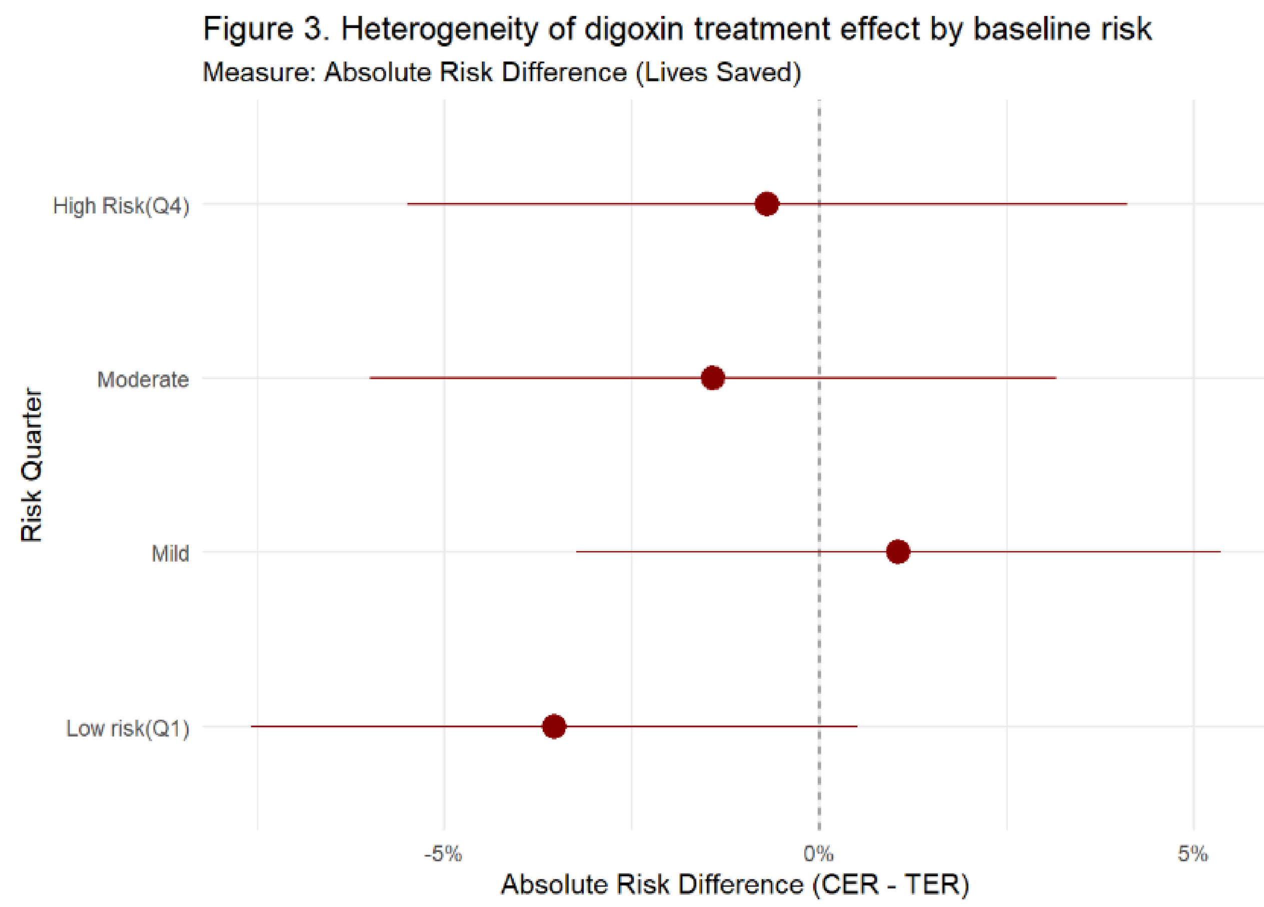

Figure 4: Forest plot of treatment heterogeneity

The solid maroon curve shows the expected absolute risk reduction under a proportional treatment effect model, illustrating increasing benefit with higher baseline risk. Blue points show observed absolute risk reductions within risk-based subgroups, with 95% confidence intervals.

Next Steps

We plan to conduct further analysis on the International Stroke Trial[5] and the National Lung screening Trial [6], to evaluate the generalizability of predictive approaches to treatment-effect heterogeneity across different clinical domain. We also plan to implement a machine-learning approach using causal forests for patient risk prediction [7].

We will use the `grf` package for this.

GitHub

The code and datasets for this project can be viewed at our GitHub repository here: https://github.com/tomdelany04/PATH-Master/

References

1. Kent et al. 2020, doi: 10.7326/M18-3668↩
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5. Sandercock et al., 2011, doi: 10.1186/1745-6215-12-101↩
6. The National Lung Screening Trial Research Team, 2011, doi: 10.1056/NEJMoa1102873↩
7. Jawadekar et al., 2023, doi: 10.1093/aje/kwad043↩